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DisBot: An Automated Medicine Dispensing Robot for Efficient Delivery

Sadad Hasan
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
1906163@eee.buet.ac.bd

Kazi Mubashir Tazwar
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
1906142@eee.buet.ac.bd

Nayeemul Islam
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
1906171@eee.buet.ac.bd

Rifah Nanzeeba
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
1906139@eee.buet.ac.bd

Md. Hasan Newaz
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
1906135@eee.buet.ac.bd

***Abstract*—Demand for efficient, accurate, and on-time delivery of medications in healthcare facilities has increased greatly in recent years to ease the burden on healthcare professionals. Moreover, prior experience related to pandemics like COVID and ever increasing risk of infections have always been a motivation to replace the manual human-based medicine delivery system with safe, accurate and machine-based automated solutions. For addressing this demand, the Automated Medicine Dispenser Robot, in short, DisBot has been designed. This robot automates the process of patient detection and medicine delivery. For guaranteed accuracy and reliability, this robot combines the realms of robotics, artificial intelligence and the IoT. Using advanced sensors, this robot can identify and sort the medicines. Additionally, using actuators it can administer the correct dose of medicine to the desired patients. It can follow patient-specific prescriptions. It significantly reduces the probability of human error, enhancing patient safety. During the design of the robot interface, user-friendliness was prioritized. Doctors, nurses and caregivers input prescription details in the database of a local host, that is, their computer. As the robot is integrated with IoT technology, it can check the updated prescription in real time and ensure drug delivery on time. As a mobile robot it can easily navigate through hospitals or nursing homes, which is vital for ensuring drug delivery. Implementation of this solution will transform the healthcare system by reducing the load on medical professionals as well as improving efficiency and minimizing errors in the medicine delivery and inventory management.**

***Index Terms*—Assistive technology, IoT in healthcare, Automated medication management, Medication error reduction, RFID**

I. Introduction

For the healthcare sector, ensuring accurate and punctual medicine delivery to the correct patients has to be one of the most challenging tasks. The challenge became tougher during the pandemic. If any error happens in medication timing and doses, that can have severe consequences. Demand for medical process automation has increased recently as the workload of healthcare professionals is growing. The DisBot is a creative and helpful solution which addresses the mentioned problems. It can also ensure increased productivity, lower human error, and improved patient outcomes. It supports efficient and reliable medicine dispensing for vulnerable patients.

The whole system has been built by integrating robotics, automation, and Internet of Things (IoT) technology . It is capable of handling any number of patients irrespective of the environment or the time. It reduces the need for human interaction with patients, making medication management swift . Inputting prescriptions becomes easier for the medical professionals because of the user-friendly interface. The user has to input the medicine names and their required doses in a spreadsheet file which contains the patients' information. The robot autonomously detects the patients by scanning RFID tags, dispenses, and delivers the appropriate medications to patients on designated locations. With real-time monitoring and updates, DisBot ensures continuous supply of medicine. At the same time it helps reduce waste. When medicine dosage changes, it does not require a complete reset of the whole system. Just updating the spreadsheet file will automatically update the DisBot's memory as it continuously accesses the spreadsheet for information. The mobility feature of the robot allows it to navigate hospital wards, clinics, or old-age homes independently, fostering convenience and accessibility. Moreover, the simplicity and efficiency of the design ensures easy maintenance, better reliability and higher acceptance. Precisely, DisBot is simple, elegant, easy to use and primed to implement solution for medicine dispensing. The Disbot is operated on a local server using a simple Microsoft Excel

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based interface.

II. Literature Review

Before commencing the initiative, thorough background study is necessary to understand the existing scenario and understand the scope for improvement and discover any research gap. Recently, automated medicine dispensers integrated into robotic systems have gained popularity, especially in hospital settings where the need for contactless medicine delivery has become critical due to the possible risk of infections. During COVID-19 like pandemics, these systems can assist medical staff to deliver medicines. Also, delivering through automated systems reduces the chances of human error.Some research works [1], [2], [5], [7]–[14], [18], [19] focused on Robotic Medicine Dispensers.Nowadays, these automated robots are designed to have the features like path-following, obstacle avoiding navigation, and conveyor systems for accurate delivery. For instance, social distancing protocols were ensured by using robotic carts as it can autonomously navigate through hospital corridors [1] [2]. Telemedicine and IoT integration enables remote monitoring and control for medical robots. Doctors can attend patients, see their medical history and prescribes through connected devices, which can be connected to the automated medicine dispensers. This ensures continuous monitoring of patients besides timely medicine delivery. So the patient's safety is increased [3] [4]. Also critical situations are addressed through emergency features, which are very thoughtful design features of these systems. Patients can use bed-side interfaces or mobile applications to alert medical staff in emergencies. The automated robotic dispensers are equipped with obstacle avoidance sensors, so it can navigate crowded environments properly and deliver the required medicines or supplies to the patients . They offer a significant advantage in epidemics or emergency situations. Their ability to operate on their own makes them an excellent solution for managing such high-stress scenarios [2] [5]. One of the key advantages of robotic medicine dispensers in hospitals is their scalability. These systems can handle a large number of patients simultaneously. During the healthcare crisis these systems become very helpful. They also minimize the risk of errors in medication delivery by automating the process. It ensures that patients don't get the incorrect medication at the wrong time. Infection control in hospital environments can be obtained by adopting contactless operations [4] [6]. A low cost automated dispenser was proposed by Rahman et al. [7] where a medicine notification and dispensing system with biometric security aims to enable physically disable patient to take medicine on time. Automated drug dispensing systems (ADDSs) have been used in the works of M. Yuan et.al. [10] [11]. However, these systems are not easy to understand and it surely takes effort for learning how to use a new system. Moreover, they were originally designed for pharmacies. Several studies [4], [15]–[17], [20] investigated RFID/IoT Healthcare Integration. While a number of papers [4], [15]–[17], [20] addressed RFID Use in Healthcare, various works [3], [4], [7]–[9], [12], [13], [15]–[17], [19], [20] explored IoT in Healthcare. Whereas, the research works [1], [2], [5], [6], [12]–[14], [19] dealt with autonomous navigation.

While some studies implemented the use of IoT and RFID in Dispensing Robots [12]–[14], others [15]–[17] reviewed the feasibility and usability of RFID based systems in healthcare sector. Additionally some research works [18], [19] tested the efficiency and practicality of robot-based systems in medicine dispensing and delivery. Patil et.al. [20] integrated RFID based systems with IoT for health monitoring purposes.

Thus, after carefully reviewing the latest efforts, the goal and the objectives were set, that is, to build up a simple, reliable, cost-effective and easy to use robot which does not require its users to use any additional application or Operating System and does not involve the usage of any unnecessary sophisticated or high-performance hardware with extraneous computational capability. Our extended analysis led us to use IoT and RFID based system for designing our efficient and reliable robot.

III. Methodology

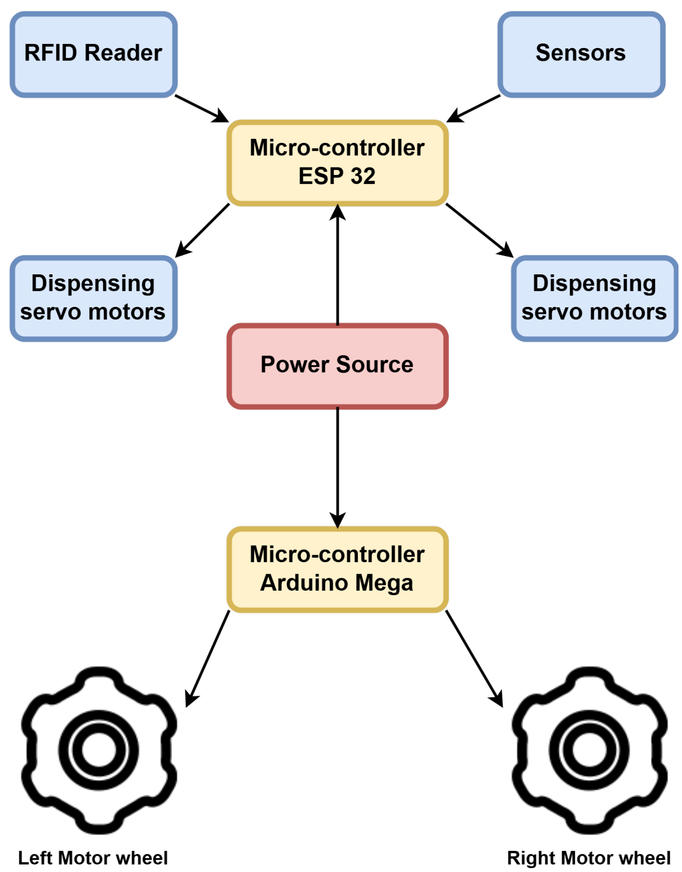

Fig. 1. Framework of DisBot

The Automated Medicine Dispenser Robot employs a multifaceted approach combining robotics, electronics, and data integration:

A. Navigation System:

For movement within hospital rooms and corridors, the robot utilizes pre-mapped routes. We have used ultrasonic sensors to detect obstacles in real-time, enabling the robot to have safe and smooth navigation.For our purpose, the hospital beds are our target. As the beds are static, a pre-mapped route was enough in this scenario.

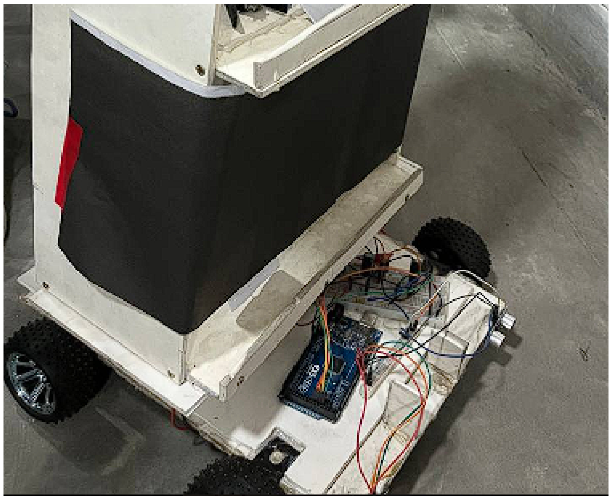

Fig. 2. Circuit for Navigation and Movement

B. Patient Identification:

The robots contain an RFID reader and patient beds have RFID tags. Each RFID is linked to a specific patient. This enables accurate patient identification. As all patient data is stored on a local server which our robot uses, it ensures delivery of correct medicine.

C. Medicine Dispensing Mechanism:

A combination of rotary movement and slide mechanism is adopted for our purpose. It is connected to the local server's spreadsheet for patient prescriptions. It ensures correct doses of correct medicine to the patients.

D. Centralized Information Retrieval:

A micro-controller accesses the data from a local server. Patient information is stored on this local server in a spreadsheet file. This server is actually a computer. So, the micro-controller and the computer should be under the same network to act flawlessly. As it can read the data during operation, it ensures it is following up to date instructions specific for the patients.

E. Hardware Components:

Main hardware for this project are servo motors, ESP32, Arduino Mega, RFID sensors and a 2200 mAh LiPo Battery. All of these components are stored in a structure of PVC boards.

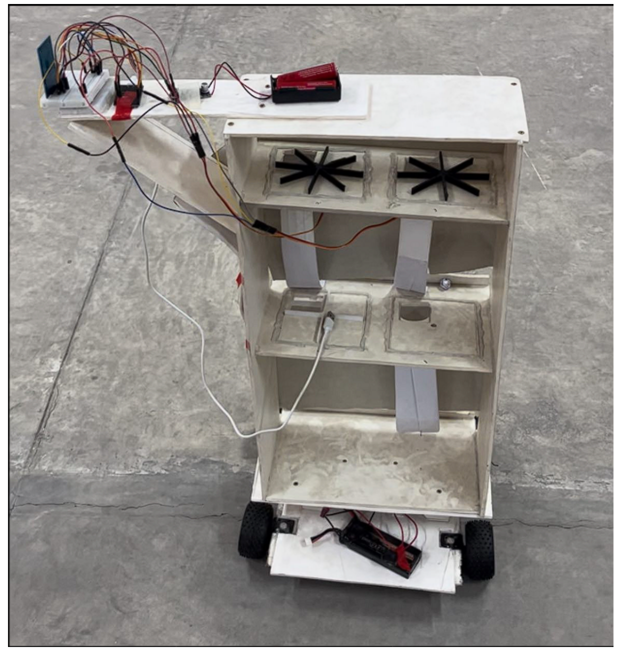

Fig. 3. The Whole Setup of DisBot

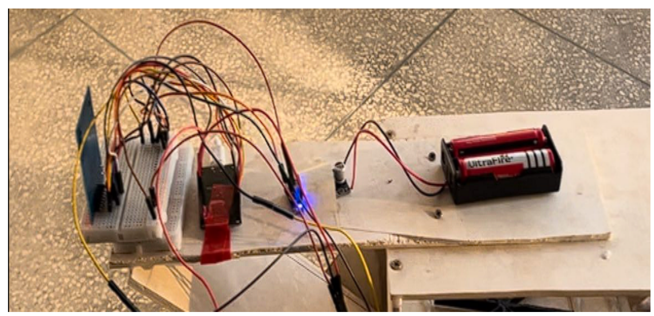

Fig. 4. RFID Card Detection Circuit

F. User Interface:

We have tried a simple interface which is intuitive. As a result, hospital staff can operate the system without extensive training.The interface have options for updating medicine schedule and monitor dispenser activity. The system uses Excel sheet to communicate the medicine dosage to DisBot. There is absolutely no requirement to operate a different software or application to convey commands to DisBot. Only simply updating the Excel sheet will send appropriate information to DisBot. For testing our setup, we kept three patient beds and a count of four medicines. The configuration is given in the Table-I.

However, the patient beds and the medicine number can easily be increased by simple modifications to the DisBot and our computer system. That is why, there is no issue with scalability and our solution can be easily transferred from one setup to another and from one location to another after incorporating some minor hardware and basic software adjustments.

TABLE I
MEDICINE ALLOCATION BY BED

Bed No	RFID No.	Med-1	Med-2	Med-3	Med-4
1	9C088ZF7	1	0	0	0
2	5CF592A1	1	0	1	1
3	AA102F987	0	1	1	1

IV. ALGORITHM

A. Robot Movement

Robot Movement Sequence

```
Initialize:
Set motor pins for forward and backward directions.
Define motor control functions: moveForward(), turnLeft(), turnRight(),stopMotors().
Set motor speed: motorSpeed = 40.
Setup:
Configure motor pins as output.
Start serial communication.
Main Steps in the Loop:
Move forward using moveForward().
Stop on RFID detection using stopMotors().
Wait 10 seconds for medicine dispensing.
Move forward again and search for RFID using moveForward().
Stop on corner detection using stopMotors().
Turn left if right side is blocked using turnLeft().
Turn right if left side is blocked using turnRight().
Execution:
The sequence runs as many times as it is programmed, stopping the robot permanently after completion.
```

B. RFID-Based Medicine Dispensing System

Medicine Dispensing System Using RFID and WiFi

```
START
Initialize serial communication at 115200 baud rate.
Initialize SPI and RFID modules.
Attach servos to pins 12, 14, 25, and 26 (set to 0 degrees).

Connect to WiFi (retry until connected).
PRINT "Connected to WiFi".
LOOP:
if new RFID card detected then
   Read and convert UID to HEX.
   PRINT "RFID UID: " + UID.
   if WiFi connected then
      Send GET request to server.
      if response received then
         Parse JSON response.
         if UID match found then
            PRINT "Match found!".
            for each medicine (1 to 4) do
               if i = 1 to medicine then
                  for i = 1 to medicine do
                     ROTATE servo to i × 45 degrees
                     PRINT "Dispensing".
                     WAIT 0.2 second.
                  end for
                  WAIT 1 second.
                  RESET servo (if needed).
               else
                  PRINT "No action".
               end if
               WAIT 2 seconds.
            end for
         else
            PRINT "No match found".
         end if
      else
         PRINT "Request failed or server offline".
      end if
   else
      PRINT "WiFi Disconnected. Reconnecting...".
   end if
else
   WAIT 1 second.
end if
END LOOP
```

V. RESULTS

A. Successful stoppage of DisBot for RFID reading

The testing has been done considering three targets (patient beds). Two beds were on row and the other was on another row. The bot followed a straight path upto a point, took a U-turn and went back. Before the U-turn two beds were to be detected and after the U-turn one bed was to be detected following the layout and floor-mapping.

TABLE II
RFID DETECTION SCORE

Patient Bed	Test Count	Detection Count	Accuracy(%)
1	20	20	100
2	20	19	95
3	20	17	85

The accuracy for Patient Bed-3 is significantly lower as the DisBot sometimes gets misaligned from its track while taking a U-turn.

B. Task Completion Time of DisBot

Each task had fixed time for completion. For example, moving forward, taking the U-turn, reading the RFID card, rotating the dispenser servo motors. It was seen that the task completion time was almost identical with the programmed time. In detail, the response time was a bit more than the programmed time. This is expected as practical scenarios are supposed to be different.

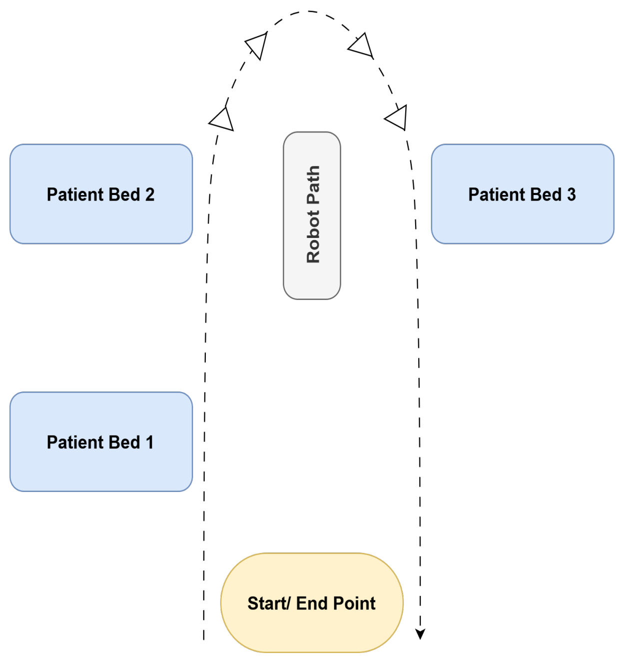

Fig. 5. Robot Pathway

VI. Comparison with existing works

Disbot successfully addresses significant deficiencies in recent research in the development of automated drug delivery systems. First of all, Haider et al. [1] developed a robotic platform to assist isolated COVID-19 patients, but their work did not specifically focus on medication dispensing. Similarly, R. Murai et al. [2] proposed a visible light communication system to facilitate the control of autonomous delivery robots in hospitals, but they did not address automation of medication dispensing. Wootton [3] investigated telemedicine trends; however, they did not cover automation in dispensing mechanisms

Many existing systems, such as those by Rahman et al. (2022) [7] and Sharmin et al. (2022) [8], lack real-time prescription updates or RFID integration. DisBot effectively solves this issue by allowing real-time prescription update.

Some systems like those by Yuan et al. (2023) [10] fail to prioritize user-friendliness and ease of adaptation. This could hinder practical deployment in real-world scenarios. DisBot requiring zero to minimal training for it to be operated can be seamlessly implemented in high-stress environments like Intensive Care Unit (ICU) or High Dependency Unit (HDU).

Except some research work /citeb9, most solutions lack scalability whereas DisBot allows easy scalability. The works of Jeon and Lee (2017) [5], Rahman et al. (2022) [7] and Sharmin et al. [8] focuses on optimizing robot scheduling without prioritizing medication accuracy at the time of dispensing. Medication accuracy is crucial as lack of medicine can disrupt recovery and delivery of wrong medicine can cause seious complications. DisBot ensures accurate delivery through its RFID-based identification system. Excluding Samson (1995) [6] and Jeon and Lee (2017) [5], all studies lack the ability to navigate through crowded places. DisBot utilizes obstacle-avoiding sensors for obstacle detection, eliminating the need of computationally expensive deep learning based obstacle detection systems which are hard to implement in mobile and embedded-system based robots.

In healthcare, data privacy is a big concern as medical data is considered to be highly sensitive. In systems like Yuan et al. (2023) [10] and Budhvant et al. (2021) [14] sufficient focus on data protection or secure data transfer is absent. As DisBot uses local servers for data storage and retrieval, the chance of compromise of data is significantly lower compared to the cloud-based systems. Due to the need for resource optimization and cost-effectiveness, DisBot came into existence. Utilizing off-the-shelf components like Arduino, ESP32, and RFID sensors, implementation cost was reduced to a great extent. The setup of DisBot can be easily completed within 6000 BDT, that is, approximately 50 USD and this equipment cost can be significantly lowered if mass production is done.

VII. Limitations and Future Works

- Interruption of operation can occur due to cable wear and tear or wire disconnections. If a concealed and compact system can be designed which can protect the delicate hardware, then this issue will be solved.Designing a more compact, robust and mechanically stable system is technically possible in this case given enough resource, time and efforts.
- In a large environment, accurate patient identification of patients might face significant challenges because of the lower range of RFID sensors. Using other methods for detecting patients to eliminate range-related limitations. This can also be done by removing the RFID system with a pre-mapped location of Bed ID where the dispensing of medicine will be linked to the patients' beds not the patients themselves. Alternatively, chipless RFID [16] can improve scalability and resolve our challenges.
- Using pre-mapped static navigation will restrict its adaptability to dynamic environments. Implementation of dynamic mapping can ensure adaptability to various environments. This can be done through robot vision. Utilizing cameras, LiDAR, infrared sensors, depth sensors, etc. For real-time object identification, pre-trained deep learning models like YOLO can be implemented. Applying Depth CNN for distance estimation of objects and Point Cloud Mapping for map construction can make the system dynamic. However, all of this comes at a computational cost. Since the objective is to design a cost-effective and reliable system, the idea of dynamic mapping was omitted. Additionally, the hospital wards are more or less static as beds, furniture and other objects are not prone to frequent relocation.
- If the patient-medicine information server and the microcontroller are not under the same network, it may not work reliably. In contrast, this feature allows the patient data to be secured and privacy is ensured as it is less prone to data breach. End-to-end encryption and blockchain technology can further increase the data security. A web-

based system can be designed using existing services of tech companies. But it should not be designed without ensuring the privacy and safety of patient information through advanced security protocols at first.
- A self-rechargeable station can be constructed where the robot can charge its batteries during idle time. This station shall be kept the base location of the robot from where it starts its operation and to which it returns upon completion.

VIII. Conclusion

This automated robot, that is, DisBot is a practical and innovative solution for an underestimated problem. It is capable of automating the medicine delivery in the healthcare system. It proves to be crucial during a pandemic. It reduces human physical interaction, which results in controlling disease transmission. The modular and scalable design principles makes it easy to incorporate this technology in various environments. Compared to contemporary solutions, DisBot is user-friendly, cost-effective, and a ready-to-use solution in practical scenarios. However, DisBot has a few limitations, but with designing iterations, it can achieve proposed modifications to address the limitations. Few considerable revisions of design will lead to a consumer level prototype. It has huge potential to automate the healthcare system making it more efficient.

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